BRETT BRINKHOFF

Senior Full Stack Engineer | TypeScript • Node.js • NestJS • React • Distributed Systems

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SUMMARY

Senior Full Stack Engineer with 15+ years building and modernizing web applications and backend services. Most recently owned the financial billing platform for a healthcare cost-sharing organization, replacing legacy .NET and Redux with TypeScript, NestJS, React, and PostgreSQL. Experienced in event driven distributed systems (RabbitMQ), contract-first API design, and mentoring across mixed-seniority teams. Applies AI coding tools (Cursor, Claude, GitHub Copilot) under team-wide standards for quality and consistency.

TECHNICAL SKILLS

- **Frontend:** TypeScript, React, Next.js, Vite, Material UI (Data Grid, X-Charts), UI virtualization, client-side state management
- **Backend:** Node.js, NestJS, REST APIs, OpenAPI/Swagger, microservices, dependency injection, Zod, class-validator, JWT/OAuth2, RBAC
- **Messaging and Cloud:** RabbitMQ (pub/sub, dead-letter queues, retries), event driven architecture, AWS (S3, SQS, SNS), Azure Entra ID, Docker, GitLab CI/CD, Doppler
- **Data:** PostgreSQL, SQL Server, Prisma, Sequelize, T-SQL, stored procedures, recursive queries, dynamic pivots, index optimization
- **Testing and Observability:** Vitest, Jest, snapshot testing, New Relic, structured logging
- **AI-Assisted Development:** Cursor, Claude, GitHub Copilot, team rule sets and prompt standards
- **Practices:** Agile/Scrum, code review, pair programming, pnpm polyrepos / monorepos
- **Earlier Career:** C#, ASP.NET MVC, .NET

PROFESSIONAL EXPERIENCE

Medi-Share (Consumer Healthcare Platform)

Senior Software Engineer (Technical Lead) *07/2017 - 06/2026*

Senior engineer on the 14-person team (9 developers, manager, scrum master, 2 CREs, DBA) that owns billing for Medi-Share, a 350,000+ member organization sharing hundreds of millions of dollars in medical bills annually.

- Created coding standards for the 9-developer team; led code reviews, paired with junior and senior engineers, and shared practices with the lead developers of two partner teams.
- Led a multi-year migration replacing legacy .NET dependencies with TypeScript microservices (NestJS, PostgreSQL, Prisma) across 7+ projects, retiring 60+ legacy endpoints. Established contract-first APIs, DTO validation (Zod and class-validator input), and discriminated-union responses, reducing production defects in billing workflows.
- Removed Redux from a billing tool during its upgrade to modern React state management; the simpler architecture let the app grow from one tool into six billing tools. Built virtualized Material UI Data Grid and X-Charts views so staff can review large sharing-fund transaction sets quickly.
- Built RabbitMQ workflows (pub/sub routing, dead-letter queues, exponential-backoff retries) so large multi-source financial queries run without blocking the API and no messages are lost; used Docker to reproduce the message flow in local development.
- Designed server-side pagination, indexing, and query optimizations for heavy PostgreSQL datasets, with scope-based RBAC and New Relic telemetry.
- Consolidated 7+ repositories into a GitLab pnpm polyrepo across DEV, TEST, STAGE, and PROD, cutting CI/CD build times by 40%+.
- Authored the team's Cursor rules (.cursorrules) and prompt guidelines, encoding architecture, type-safety, and testing standards so AI-assisted code follows team conventions.
- Delivered projects using Next.js, Vite, Vitest and Jest, Doppler secrets management, and Azure Entra ID authentication.
- Earlier in tenure, led a dedicated team that built a claims cost-management platform (ICD-10, CPT, HCPCS) tracking every major claim above a set dollar threshold, enabling claims staff to negotiate early-payment options with providers and saving millions of dollars.

American Fidelity (contract via TEKsystems)
Senior Software Engineering Consultant *02/2017 - 07/2017*

- Built a C#/ASP.NET probabilistic data matching and cleansing service that unified customer records across millions of rows from internal and external data feeds.
- Integrated Azure Identity and built RBAC admin tools to manage permissions across enterprise financial applications.

Compassion International (contract via TEKsystems)
Senior Software Engineering Consultant *03/2014 - 01/2017*

- Co-led a task force that grew the global communication system's annual capacity from 13M to 150M+ messages (platform processing $1B+ in annual donor transactions) using a stateful C# queuing service that buffers, batches, and polls releases to XMPie.
- Built SQL Server data pipelines (dynamic T-SQL pivots, unpivots, staging tables) to aggregate and transform large payload batches.
- Designed a REST error-logging system with local event-log fallback, adopted as the logging standard across all enterprise projects.

Rendi / Danielson Designs
Lead Software Engineer *06/2010 - 03/2014*

- Lead programmer for the small in-house team that built all of the company's business software for Rendi, an Otterbox-funded maker of custom-designed products.
- Proposed and delivered an in-house multi-tier sales-representative commission and payment system, replacing a $15K/month third-party platform (ASP.NET Web Forms, Visual Basic, SQL Server), launched in 6 months, followed by 6 months of enhancements.
- Built the customer site with a custom shopping cart for personalized products, PCI-compliant checkout with multiple payment options, and Facebook login, plus a jQuery product designer (responsive on mobile) over a catalog of hundreds of products with millions of customization combinations.
- Built the fulfillment system with barcode scanning and per-item tracking through shipping, integrating ProPay, FedEx, UPS, USPS, and Avalara tax services via REST and SOAP; used AWS S3 for large-scale image storage and SQS/SNS for text messaging to stylists and employees.

EDUCATION

Bachelor of Science (B.S.) | University of Nevada, Las Vegas

- NCAA Division I All-Conference Diver & Academic All-American